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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	:	**B.E :- Common to CSE / ISE / AI & ML / AI & DS**	**Semester**	:	**III**
Course Name	:	**Linear Algebra and Laplace Transforms**	**Max. Marks**	:	**100**
Course Code	:	**CS / IS / AI / AD31**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) Find (i) $L\{te^{-t}\sin 3t\}$ (ii) $L\left\{\frac{1-e^{t}}{t}\right\}$. CO1 (06)

b) Find the Laplace transform of the periodic function of period 2π is defined by $f(t)=\begin{cases} t, & 0<t<\pi \\ \pi-t, & \pi<t<2\pi \end{cases}$. CO1 (07)

c) Evaluate $\int_0^{\infty} t\, e^{-2t}\cos 3t\, dt$. CO1 (07)

2. a) Show that $L\{\sin\sqrt{t}\}=\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}e^{-\frac{1}{4s}}$. CO1 (06)

b) If $L\{f(t)\}=F(s)$ then prove that $L\{(\sinh at)f(t)\}=\frac{1}{2}[F(s-a)-F(s+a)]$ and also evaluate $L[t^8\sinh 2t]$ CO1 (07)

c) If $L\{f(t)\}=F(s)$, then prove that $L\{t^n f(t)\}=(-1)^n\frac{d^n[F(s)]}{ds^n}$ where n is a positive integer. CO1 (07)

UNIT - II

3. a) State and prove convolution theorem. CO1 (06)

b) Express the function $f(t)=\begin{cases} 1 & 0<t\le 1 \\ t & 1<t\le 2 \\ t^2 & t>2 \end{cases}$ in terms of the Heaviside function and hence find its laplace transform. CO1 (07)

c) Solve $\frac{dx}{dt}-y=e^{-t}$, $\frac{dy}{dt}+x=Sint$, given that $x(0)=1$, $y(0)=0$ using Laplace transforms. CO1 (07)

4. a) Find the inverse Laplace transform of $log\left(\frac{s^2+9}{s(s+9)(s-9)}\right)$. CO2 (06)

b) Solve by using Laplace transforms $y''+4y'+3y=e^{-t}$ given that $y(0)=y'(0)=1$. CO2 (07)

c) Using convolution theorem find the inverse Laplace transform of $\frac{s^2}{(s^2+1)(s^2+4)}$. CO2 (07)

UNIT - III

5.	a)	Show that the sets of vectors {(1, 2, 3), (-1, -1, 0), (2, 5, 4)} span R^3. Express the vector (1, 3, -2) in terms of each spanning set.	CO3	(06)
	b)	Prove that the transformation T: $R^2 \rightarrow R^3$ given by T(x, y) = (3x+ y, 2y, x - y) is linear. Find the images of (1, 2) and (2, -5) under this transformation.	CO3	(07)
	c)	Define kernel and range of a linear transformation and use it to find the dimension of the kernel and range of the linear transformations defined by the matrix $\begin{bmatrix} 1 & -2 & 3 & 5 \\ 1 & -1 & 8 & 7 \\ 2 & -4 & 6 & 10 \end{bmatrix}$.	CO3	(07)
6.	a)	Determine the matrix that defines a reflection in the x-axis followed by a rotation through an angle $\frac{\pi}{2}$ and then by the dilation of factor 3. Find the image of the point $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ under the sequence of mappings.	CO3	(06)
	b)	State Rank and Nullity theorem and Verify Rank nullity theorem for T(x, y) = (3x, x-y, y) of $R^2 \rightarrow R^3$.	CO3	(07)
	c)	Consider the bases B = {(1, 2), (3, -1)} and B' = {(3, 1), (5, 2)} of R^2. Find the transition matrix from B to B'. If u is a vector such that $u_B = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, find $u_B{}'$.	CO3	(07)

UNIT- IV

7.	a)	Find QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.	CO4	(10)
	b)	Find a basis for null space, column space and hence the dimension of a null space and column space of a matrix $A = \begin{bmatrix} -3 & 9 & -2 & -7 \\ 2 & -6 & 4 & 8 \\ 3 & -9 & -2 & 2 \end{bmatrix}$.	CO4	(10)
8.	a)	Construct orthonormal basis for $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$, w =Span$\{x_1, x_2, x_2\} \in R^4$, using Gram-Schmidt process.	CO4	(10)
	b)	Find the least-squares solution of $Ax = b$, where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.	CO4	(10)

UNIT - V

9. a) Diagonalize the matrix $A=\begin{bmatrix}1 & -1\\ 2 & 4\end{bmatrix}$ and hence find A^6. CO5 (10)

b) Find an Singular value decompositions of the matrix $A=\begin{bmatrix}4 & 11 & 14\\ 8 & 7 & -2\end{bmatrix}$. CO5 (10)

10. a) Orthogonally diagonalize the matrix $A=\begin{bmatrix}3 & 1\\ 1 & 3\end{bmatrix}$. CO5 (10)

b) (i) Determine whether the matrix is Hermitian CO5 (10)

$$A=\begin{bmatrix}3 & 7-4i & -2+5i\\ 7+4i & -2 & 3+i\\ -2-5i & 3-i & 4\end{bmatrix}.$$

(ii) Is matrix represented by quadratic from $3x^2+5y^2+3z^2-2yz+2zx-2xy$ is positive definite?

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